

DETERMINANTS

Exercise 1

Exercise 1.a)

Let's compute $\Delta = \begin{vmatrix} x & y & x+y \\ y & x+y & x \\ x+y & x & y \end{vmatrix}$ $C_1 \leftarrow C_1 + C_2 + C_3$

$$\begin{aligned}
 &= \begin{vmatrix} 2(x+y) & y & x+y \\ 2(x+y) & x+y & x \\ 2(x+y) & x & y \end{vmatrix} \\
 &= 2(x+y) \begin{vmatrix} 1 & y & x+y \\ 1 & x+y & x \\ 1 & x & y \end{vmatrix} \quad \begin{array}{l} R_1 \leftarrow R_1 - R_3 \\ R_2 \leftarrow R_2 - R_3 \end{array} \\
 &= 2(x+y) \begin{vmatrix} 0 & y-x & x \\ 0 & y & x-y \\ 1 & x & y \end{vmatrix} \\
 &= 2(x+y) \begin{vmatrix} y-x & x \\ -y & x-y \end{vmatrix} \\
 &= 2(x+y) [-(x-y)^2 - xy] \\
 &= -2(x+y) [x^2 - xy + y^2]
 \end{aligned}$$

Let's compute:

$$\begin{aligned}
 \Delta &= \begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a+2)^2 & (b+2)^2 & (c+2)^2 \end{vmatrix} = \begin{vmatrix} a^2 & b^2 & c^2 \\ a^2+2a+1 & b^2+2b+1 & c^2+2c+1 \\ a^2+4a+4 & b^2+4b+4 & c^2+4c+4 \end{vmatrix} \\
 &\quad \begin{array}{l} R_2 \leftarrow R_2 - R_1; R_3 \leftarrow R_3 - R_1 \\ R_3 \leftarrow R_3 - 2R_2 \end{array} \\
 \Delta &= \begin{vmatrix} a^2 & b^2 & c^2 \\ 2a+1 & 2b+1 & 2c+1 \\ 4a+4 & 4b+4 & 4c+4 \end{vmatrix} \\
 &= \begin{vmatrix} a^2 & b^2 & c^2 \\ 2a+1 & 2b+1 & 2c+1 \\ 2 & 2 & 2 \end{vmatrix} \quad R_2 \leftarrow R_2 - \frac{1}{2} R_3 \\
 &= 2 \begin{vmatrix} a^2 & b^2 & c^2 \\ 2a & 2b & 2c \\ 1 & 1 & 1 \end{vmatrix} \\
 &= 4 \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix} \quad \begin{array}{l} \text{Wandermonde determinant} \\ V(a, b, c) \text{ see Bonus exercise} \end{array} \\
 &= \dots = -4(c-a)(b-a)(c-b).
 \end{aligned}$$

Exercise 1.b)

Exercise
2.a)

$$\Delta_a = \begin{vmatrix} 1 & \cos(a) & \cos(2a) \\ 1 & \cos(b) & \cos(2b) \\ 1 & \cos(c) & \cos(2c) \end{vmatrix}$$

Reminder: $\cos(2\theta) = 2\cos^2\theta - 1$

$$\Delta_a = \begin{vmatrix} 1 & \cos a & 2\cos^2 a - 1 \\ 1 & \cos b & 2\cos^2 b - 1 \\ 1 & \cos c & 2\cos^2 c - 1 \end{vmatrix}$$

$$= 2 \begin{vmatrix} 1 & \cos a & \cos^2 a \\ 1 & \cos b & \cos^2 b \\ 1 & \cos c & \cos^2 c \end{vmatrix}$$

$$= 2 \cdot V(\cos a, \cos b, \cos c) \quad \text{Vandermonde again or compute it manually...}$$

$$= \dots$$

$$= 2 (\cos b - \cos a)(\cos c - \cos a)(\cos c - \cos b).$$

Exercise
3.a)

$$\Delta_x = \begin{vmatrix} 1 & 1 & 2 & 3 \\ 1 & 2-x^2 & 2 & 3 \\ 2 & 3 & 1 & 5 \\ 2 & 3 & 1 & 9-x^2 \end{vmatrix}$$

What about x ??

- If $x = 1$ then the first two rows are the same; so $\Delta_x = 0$

($\Delta_x = 0 \Rightarrow$ Reminder: the matrix is not invertible; some rows (or cols) can be written as a linear combination of the other rows (or cols) resp.)

- If $x = 2$ then the two last rows are the same, so $\Delta_x = 0$.

$$\Delta_x = \begin{vmatrix} 1 & 1 & 2 & 3 \\ 0 & 1-x^2 & 0 & 0 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 4-x^2 \end{vmatrix} \quad \begin{array}{l} L_2 \leftarrow L_2 - L_1 \\ L_3 \leftarrow L_3 - 2L_1 \\ L_4 \leftarrow L_4 - L_3 \text{ (before changing } L_3) \end{array}$$

$$= \begin{vmatrix} 1-x^2 & 0 & 0 \\ 1 & -3 & -1 \\ 0 & 0 & 4-x^2 \end{vmatrix}$$

$$= (4-x^2) \begin{vmatrix} 1-x^2 & 0 \\ 1 & -3 \end{vmatrix}$$

$$= (4-x^2) [-3(1-x^2) - 0]$$

$$= -3(4-x^2)(1-x^2) \quad \text{ok with cases } x=1 \text{ and } x=2.$$

LINEAR MAP / DETERMINANT

Exercise 4

$$T: M_n(\mathbb{R}) \rightarrow M_n(\mathbb{R})$$

$$M \mapsto {}^t M$$

We want $\det(T)$, i.e.

NB: another notation for the transpose

What is the determinant of the matrix representation of the linear application T in the standard basis of $M_n(\mathbb{R})$?

Remark: $T \circ T = \text{Id}$

$$\text{so } \det(T)^2 = 1$$

But: we are looking for $\det(T)$

♡ Step 1: Prove: T is a linear application

Practice
at home

♡ Step 2: let us denote by $S_n(\mathbb{R})$ the set of symmetric matrices of $M_n(\mathbb{R})$; $A_n(\mathbb{R})$ the set of antisymmetric matrices of $M_n(\mathbb{R})$.

- Show that $A_n(\mathbb{R})$ and $S_n(\mathbb{R})$ are subspaces of $M_n(\mathbb{R})$
- Show $M_n(\mathbb{R}) = S_n(\mathbb{R}) \oplus A_n(\mathbb{R})$ (direct sum).
- Find a basis of $S_n(\mathbb{R})$ and a basis of $A_n(\mathbb{R})$.

$$\text{Deduce that } \dim(S_n(\mathbb{R})) = \frac{n(n+1)}{2}$$

$$\text{and } \dim(A_n(\mathbb{R})) = \frac{n(n-1)}{2}.$$

Hint: consider the standard basis of $M_n(\mathbb{R})$. We denote its vectors E_{ij} , with the coefficients null except the coefficient at row i and column j , equal to 1. Consider the family of vectors $\{E_{ii}, i \in \llbracket 1, n \rrbracket\} \cup \{E_{ij} + E_{ji}, i \neq j, i, j \in \llbracket 1, n \rrbracket\}$ and $\{E_{ij} - E_{ji}, i \neq j, i, j \in \llbracket 1, n \rrbracket\}$.

Bonus Exercise

Vandermonde

Bonus exercise: Vandermonde determinant.

Let $(a_1, \dots, a_n) \in \mathbb{R}^n$.

$$\text{Let } V(a_1, \dots, a_n) = \det \begin{pmatrix} 1 & 1 & \dots & 1 \\ a_1 & a_2 & \dots & a_n \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{n-1} & a_2^{n-1} & \dots & a_n^{n-1} \end{pmatrix}$$

Prove by induction that $V(a_1, \dots, a_n) = \prod_{1 \leq i < j \leq n} (a_j - a_i)$ (*)

NB: you can consider that $a_1 \neq a_2 \neq \dots \neq a_n$.

Hint: consider the polynomial $P(x) = V(a_1, a_2, \dots, a_{n-1}, \frac{x}{a_n - a_{n-1}})$ instead of a_n .

Proof sketch

Degree of $P \leq n-1$

$$P(a_1) = P(a_2) = \dots = P(a_{n-1}) = 0$$

so $P(x) = \lambda \prod_{i=1}^{n-1} (x - a_i)$ where λ is the leading coefficient of the polynomial;

i.e. the coefficient of x^{n-1} .

$$\text{Hence } \lambda = \det \begin{pmatrix} 1 & 1 & \dots & 1 \\ a_1 & a_2 & \dots & a_{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{n-2} & a_2^{n-2} & \dots & a_{n-1}^{n-2} \end{pmatrix} = V(a_1, \dots, a_{n-1}).$$

$$\text{So we get } V(a_1, \dots, a_n) = \prod_{i=1}^{n-1} (a_n - a_i) V(a_1, \dots, a_{n-1}) (*)$$

• Case $n=2$

$$V(a_1, a_2) = \begin{vmatrix} 1 & 1 \\ a_1 & a_2 \end{vmatrix} = a_2 - a_1 = (a_2 - a_1) \times (1) \quad \text{ok}$$

• Suppose (*) is true for a given $n \geq 2$.

Using (*) we prove that it holds for $n+1$.

• Conclusion.

Bonus Exercise

Vandermonde (detailed)

Bonus exercise: Vandermonde determinant.

Let $(a_1, \dots, a_n) \in \mathbb{R}^n$.

$$\text{Let } V(a_1, \dots, a_n) = \det \begin{pmatrix} 1 & 1 & \dots & 1 \\ a_1 & a_2 & \dots & a_n \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{n-1} & a_2^{n-1} & \dots & a_n^{n-1} \end{pmatrix}$$

Prove by induction that $V(a_1, \dots, a_n) = \prod_{1 \leq i < j \leq n} (a_j - a_i)$ (*)

NB: you can consider that $a_1 \neq a_2 \neq \dots \neq a_n$ because otherwise, we directly have $V(a_1, \dots, a_n) = 0$ (repeated column) and the product is null too ($a_j - a_i$).

Hint: consider the polynomial $P(x) = V(a_1, a_2, \dots, a_{n-1}, x)$. This polynomial is obtained by "replacing a_n by x " instead of a_n .

With details

Degree of $P \leq n-1$ since we elevate x to the power $(n-1)$ by definition of $V(a_1, \dots, a_{n-1}, x)$.

$$P(a_1) = P(a_2) = \dots = P(a_{n-1}) = 0$$

so $P(x) = \lambda \prod_{i=1}^{n-1} (x - a_i)$ where λ is the leading coefficient of the polynomial;

writing P in a factorized manner

i.e. the coefficient of x^{n-1} .

$$\text{Hence } \lambda = \det \begin{pmatrix} 1 & 1 & \dots & 1 \\ a_1 & a_2 & \dots & a_{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{n-2} & a_2^{n-2} & \dots & a_{n-1}^{n-2} \end{pmatrix} = V(a_1, \dots, a_{n-1}) \quad (*)$$

$$\text{So we get } V(a_1, \dots, a_n) = \left(\prod_{i=1}^{n-1} (a_n - a_i) \right) V(a_1, \dots, a_{n-1}) \quad (*)$$

by substituting x with a_n in the polynomial

the factorization using the roots

the leading coefficient

• Case $n=2$

$$V(a_1, a_2) = \begin{vmatrix} 1 & 1 \\ a_1 & a_2 \end{vmatrix} = a_2 - a_1 = (a_2 - a_1) \times 1 \quad \text{ok}$$

• Suppose (*) is true for a given $n \geq 2$.

Using (*) we prove that it holds for $n+1$.

• Conclusion.

(*) Explanation:

$$V(a_1, \dots, a_{n-1}, x) = \det$$

$$\begin{bmatrix} 1 & 1 & \dots & 1 \\ a_1 & a_2 & \dots & a_{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{n-2} & a_2^{n-2} & \dots & a_{n-1}^{n-2} \\ a_1^{n-1} & a_2^{n-1} & \dots & a_{n-1}^{n-1} \end{bmatrix} \begin{matrix} 1 \\ x \\ \vdots \\ x^{n-2} \\ x^{n-1} \end{matrix}$$

when applying the expansion formula on the last column, we see that the cofactor associated to each coefficient x^i of this last column corresponds to the coefficient λ_i associated with x^i in the polynomial.

In particular, we recognize that the leading coefficient λ_{n-1} (denoted λ above) is $V(a_1, \dots, a_{n-1})$.

Bonus Exercise

Vandermonde
(continued)

Proof by induction

Proving the "hereditary" property:

Reminder: (i) we have proven the property "manually" in the case $n=2$

(ii) we suppose that the property is true for a given $n \geq 2$.

i.e. for a given $n \geq 2$,

$$V(a_1, \dots, a_n) = \prod_{1 \leq i < j \leq n} (a_j - a_i). \quad \bullet$$

Now we consider $V(a_1, \dots, a_n, a_{n+1})$.

Using (*) : $V(a_1, \dots, a_n, a_{n+1}) = V(a_1, \dots, a_n) \prod_{1 \leq i \leq n} (a_{n+1} - a_i)$

Using $\bullet$ we can write

$$\begin{aligned} V(a_1, \dots, a_n, a_{n+1}) &= \prod_{1 \leq i < j \leq n} (a_j - a_i) \prod_{1 \leq i \leq n} (a_{n+1} - a_i) \\ &= \prod_{1 \leq i < j \leq \underline{n+1}} (a_j - a_i) \end{aligned}$$

So the property is true for $n+1$.

(iii) Finally, we conclude that the property is true for all $n \geq 2$.

Bonus Exercise: Block matrix

We consider the bloc matrix M

$$M = \left(\begin{array}{c|c} A & B \\ \hline O & C \end{array} \right)$$

$O \in M_{p,n}(\mathbb{R})$

General case $\rightarrow$ We work with square matrices here

$$\left\{ \begin{array}{l} A \in M_{p,q}(\mathbb{R}) \\ B \in M_{p,n}(\mathbb{R}) \\ C \in M_{m,n}(\mathbb{R}) \end{array} \right. \left(\begin{array}{l} A \in M_{n,n}(\mathbb{R}) \\ B \in M_{n,p}(\mathbb{R}) \\ C \in M_{pp}(\mathbb{R}) \end{array} \right)$$

"O" is the matrix in $M_{m,q}(\mathbb{R})$ with all coefficients = zeros.

Prove that $\det(M) = \det(A) \times \det(C)$.

Case 1: Suppose $\det(A) = 0$.

This means that there exist a column $[A]_j$ of A that can be expressed as a linear combination of the other columns of A .

Since all the columns of $O_{m,q}$ are identical (filled with zeros), the column j of M can be expressed as a linear combination of the other first q columns of M (with the same weights as the previous linear combination for $[A]_j$).

So $\det(M) = 0$ and indeed $\det(M) = \det(A) \det(C)$.

Case 2 Suppose $\det(A) \neq 0$. The inverse of A exists.

We remark that M can be written as:

$$M = \left(\begin{array}{c|c} A & O \\ \hline O & I_p \end{array} \right) \left(\begin{array}{c|c} I_n & A^{-1}B \\ \hline O & C \end{array} \right), \text{ and } \det(M) = \det \left(\begin{array}{c|c} A & O \\ \hline O & I_p \end{array} \right) \det \left(\begin{array}{c|c} I_n & A^{-1}B \\ \hline O & C \end{array} \right)$$

by expanding along the last column, then for last column etc. p times, we obtain $\det(A)$

by expanding along the first column, then second column... n times, we obtain $\det(C)$

So $\det(M) = \det(A) \det(C)$.